HYPOXIA : DECODED AT MOLECULAR LEVEL

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ABSTRACT

Oxygen is essential for animal life. It is used by the mitochondria, virtually present in all animal cells to convert food into useful energy. Our body responds differently under varied oxygen conditions. Through evolution, our body has developed mechanisms to ensure sufficient supply of oxygen. A key physiological response to hypoxia (low oxygen level) is the rise in the hormonal levels of erythropoietin (EPO). EPO leads to an increased level of red blood cells through erythropoiesis. The EPO gene is found to be present in all tissues. This shows that the mechanism is general and functional in many different cell types and not only restricted to kidney cells where EPO is generally produced. It has been shown that this mechanism is oxygen-dependent. The EPO gene works when a protein complex binds to the identified DNA segment. This protein complex is known as Hypoxia Inducible Factor (HIF) that consists of two different DNA binding proteins; HIF-1α and Aryl Hydrocarbon Receptor Nuclear Translocator (ARNT). At normal oxygen levels, a cellular machinery, proteasome degrades HIF-1α. HIF-1α is tagged by ubiquitin for degradation. Von Hippel Lindau (VHL) is a part of complex that labels proteins with ubiquitin. This demonstrates that VHL can physically interact with HIF-1α and is required for its degradation at normal oxygen levels. Under normoxia (normal) conditions, hydroxyl groups are added at two specific positions in HIF-1α; in a process known as prolyl hydroxylation. This allows VHL to recognize and bind to HIF-1α. When oxygen levels are low (hypoxia), HIF-1α is protected from degradation and accumulates in the nucleus where it associates with ARNT and binds to specific DNA sequences in hypoxia regulated genes. So there is increase in EPO hormone which accelerates the process of erythropoeisis. Oxygen sensing by cells allows them to adjust their metabolism to low oxygen levels, like in the muscles during intense exercise. Other examples include the generation of new blood vessels and the production of red blood cells which is essential for fetal and placental development. Oxygen sensing is central to a large number of diseases as in patients with chronic renal failure suffering from anaemia. Moreover oxygen-regulated machinery has an important role in cancer. In tumors, the oxygen regulated machinery is utilized to stimulate blood cell formation and reshape metabolism for effective proliferation of cancer cells. Intense ongoing efforts in academic laboratories and pharmaceutical companies are now focused on developing drugs that can interfere with different diseased states by either activating or blocking the oxygen-sensing machinery.

OXYGEN AT CENTRE STAGE:

- Oxygen is a diatomic molecule.
- Essential for animal life.
- Used to convert food into useful energy, needed for the proper functioning and working of the body.
- Our body responds differently under varied oxygen conditions.
- Hypoxia is a condition where the oxygen levels are low than usual.
- In such condition, body releases EPO (erythropoietin) hormone which stimulates red blood cell production to maintain the oxygen homeostasis in the body.

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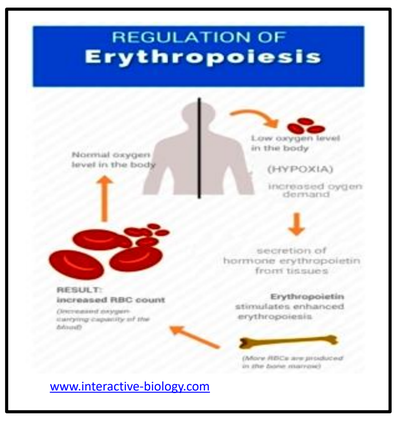
CONDITIONS IN BODY:

- Under normal oxygen conditions (Normoxia), less production of hormone erythropoietin leads to the decreased RBCs count.
- Under less oxygen conditions (Hypoxia), the increased production of hormone erythropoietin leads to the subsequent increase in the RBC count.
- This increases the oxygen carrying capacity of the blood.

Normoxia–↓hormone EPO→↓ production of RBCs
Hypoxia –↑hormone EPO→↑production of RBCs



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WORKING AT MOLECULAR LEVEL:

- The EPO gene works when a protein complex binds the identified DNA segment i.e., the hypoxia regulated genes (HRE).
- This protein complex is known as Hypoxia Inducible Factor (HIF)
- Consists of two DNA binding proteins, HIF-1α and Aryl Hydrocarbon Receptor Nuclear Translocator

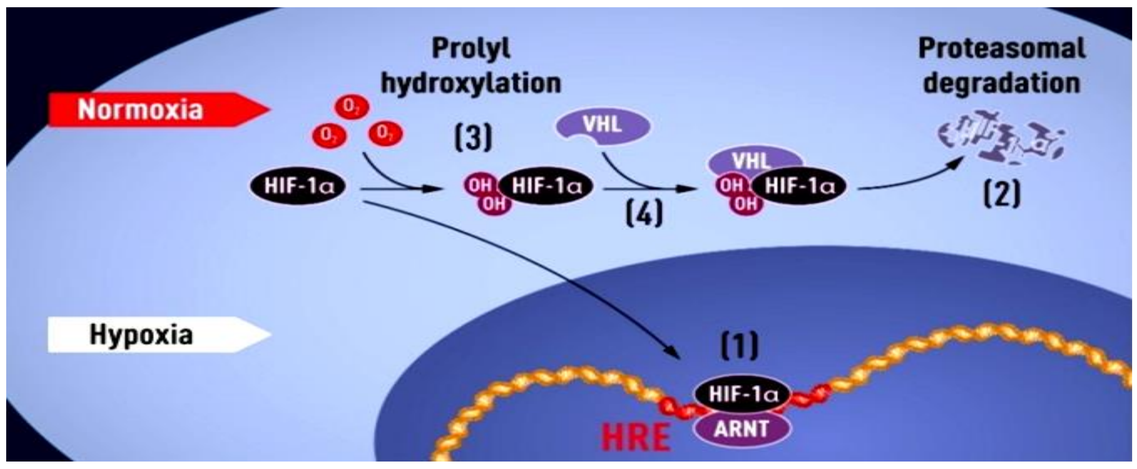

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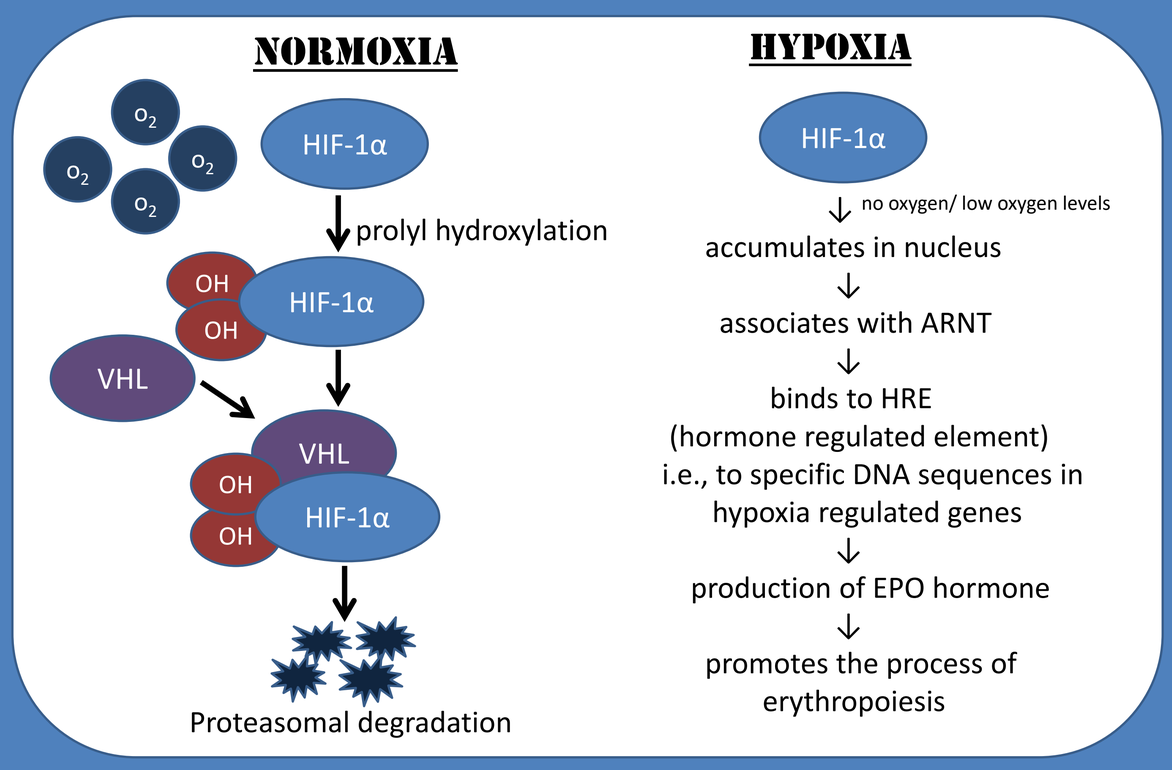
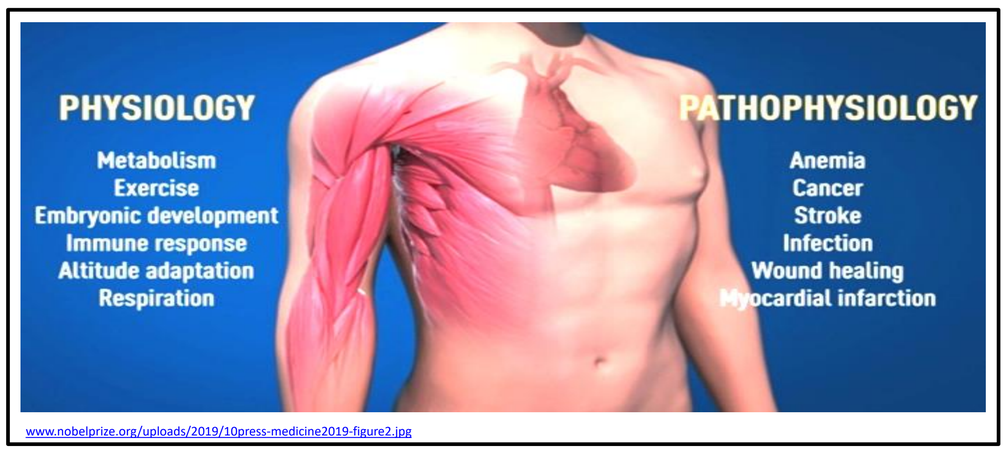
Fig.1 ·During hypoxia, the HIF-1α is protected from degradation and continues to increase. It accumulates in the nucleus where it gets integrated with ARNT and binds to HRE.[1]
·During normoxia, oxygen binds to HIF-1α by prolyl hydroxylation.[3]
·This allows VHL to recognize and bind to HIF-1α which is tagged by ubiquitin.[4]
·This accompanies HIF-1α to proteasome which recognizes it and signals for the disassembly of the HIF-1α.[2]



IMPORTANCE OF OXYGEN SENSING MACHINERY:



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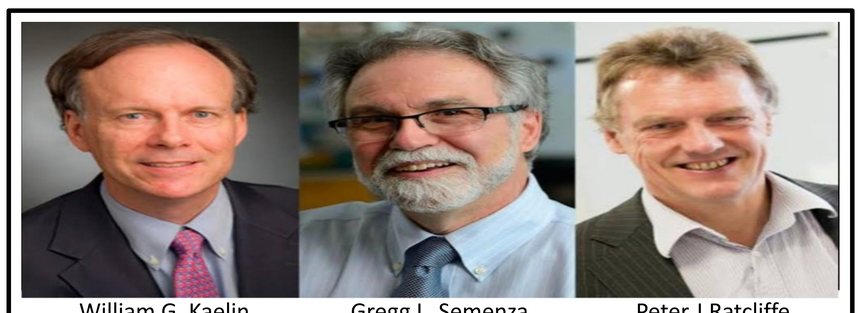
- Oxygen sensing allows cells to adapt their metabolism to low oxygen levels, for example in our muscles during intense exercise or the generation of new blood cells and the production of RBCs essential for fetal and placental development.
- Oxygen sensing is central to a large number of diseases as well as the immune response of the body.
- Moreover oxygen regulated machinery has a significant role in stimulating the blood cell formation as well as reshaping metabolism for effective proliferation of cells in case of cancers and tumors.
- It also affects many pathological processes such as wound healing which requires RBCs and platelets.
- Oxygen sensing plays a central role in many diseases such as chronic renal failure and anemia.
- Intensive efforts are ongoing in academic laboratories and pharmaceutical companies to develop new drugs to intervene and affect many diseased states by either activating or blocking the oxygen sensing machinery.

NOBEL PRIZE WINNERS FOR OXYGEN-SENSING MACHINERY



William G. Kaelin | Gregg L. Semenza | Peter J.Ratcliffe

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The Nobel Prize in Physiology or Medicine 2019 is awarded jointly to William G. Kaelin, Sir Peter J. Ratcliffe and Gregg L. Semenza "for their discoveries of how cells adapt to oxygen availability".

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